Unit 1.1 Inequalities etc (Self study)

Note Title 2015/01/25

READ: Stewart A2 – A6

We focus on inequalities only in class.

- Writing mathematics is important
- Notation and symbols need to be correct

Today:

One symbol, two words

$\Longrightarrow$ implies

and / or.

Imply: $x(x+1) = 0 \Longrightarrow x = 0$ or $x+1 = 0$

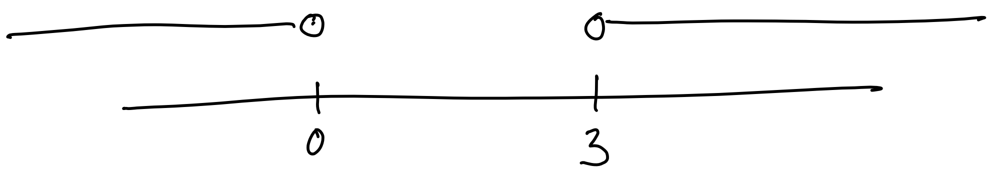
$\Longrightarrow x = 0$ or $x = -1$.

- $x > 3$ and $x < 5$

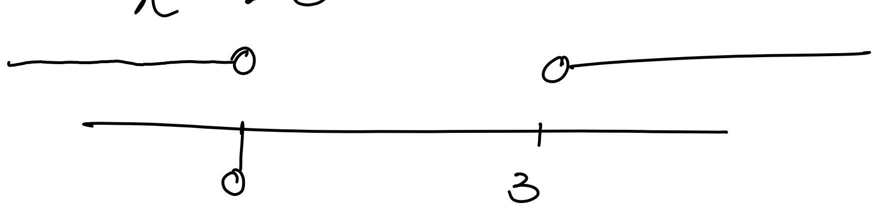
$3 < x < 5$

$\Longrightarrow x \in (3,5)$.

- $x > 3$ or $x < 0 \Rightarrow x \in (-\infty, 0) \cup (3, \infty)$.

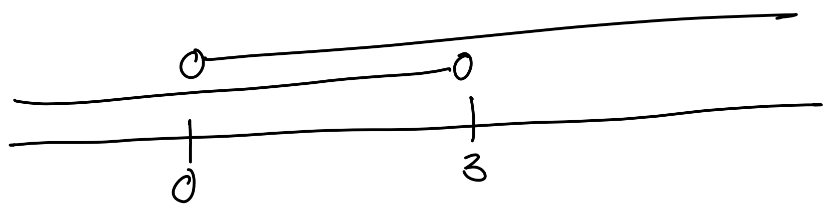


- $x > 3$ and $x < 0$



$\Rightarrow x \in \phi = \{\}$

- $x > 0$ or $x < 3$



$\Rightarrow x \in \mathbb{R}$

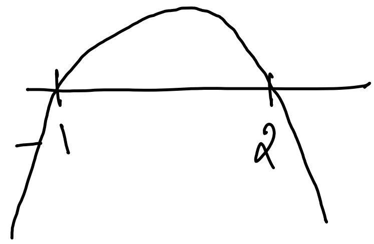
Eg: Solve for x if $\dfrac{x+1}{2-x} \geq 0$

$$\frac{x+1}{2-x} \geq 0$$

$$\Rightarrow \frac{x+1}{2-x} \times (2-x)^2 \geq 0 \times (2-x)^2 \qquad (x \neq 2)$$

$$\Rightarrow (x+1)(2-x) \geq 0$$

$$\Rightarrow x \in [-1, 2)$$



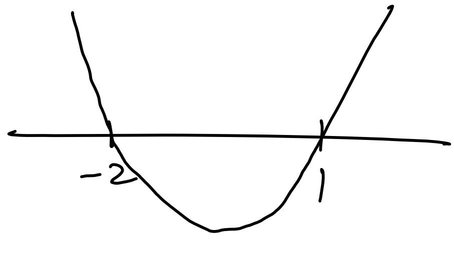
Eg. Solve for x: $x^2 + x < 2$

$$x^2 + x < 2$$

$$\Rightarrow x^2 + x - 2 < 0$$

$$\Rightarrow (x-1)(x+2) < 0$$

$$\Rightarrow x \in (-2, 1)$$



Eg. Solve for x: $\dfrac{x-1}{\underset{+}{(x-2)^2}} \geq 0 \quad +$

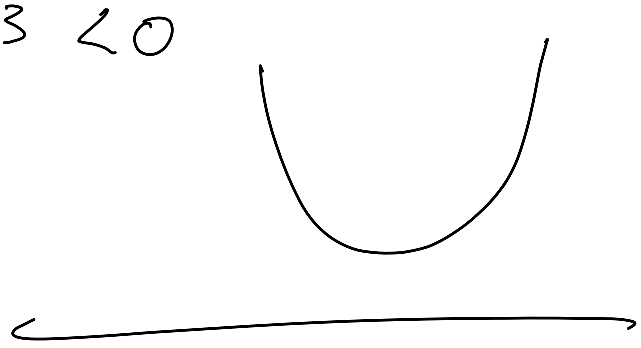
$\Rightarrow \quad x \in [1, \infty) \, , \quad x \neq 2$

$x - 1 > 0 \Rightarrow x > 1 \quad , x \neq 2$

<u>Eg.</u> Solve for x if $x^2 + x + 1 > 0$

$b^2 - 4ac = -3 < 0$

So no roots



$\underline{x \in \mathbb{R}}$.